



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

SUMMER 25-26

Software Quality Assurance and Testing

Section: C

**MediBook: Online Doctor Appointment System**

A Report submitted

To

**SAIKAT BAUL**

A Report submitted

By

| SN | Student Name       | Student ID |
|----|--------------------|------------|
| 16 | Dipaneeta Pramanik | 22-48286-3 |
| 17 | Mahin Sarker       | 22-48338-3 |
| 18 | Rayhan Chowdhury   | 22-48420-3 |
| 20 | Tanni Rani Nath    | 22-48430-3 |
| 21 | Fahim Rahman Talha | 22-48437-3 |

---

# Software Test Plan

for

## MediBook: Online Doctor Appointment System

Version 1.0 approved

Prepared by Dipaneeta Pramanik, Mahin Sarker, Rayhan Chowdhury  
Tanni Rani Nath, Fahim Rahman Talha

American International University-Bangladesh

2 September 2026

## Table of Contentes:

| Item No. | Item                                  | Pg. No. |
|----------|---------------------------------------|---------|
| 1.       | TEST PLAN IDENTIFIER.....             | 3       |
| 2.       | INTRODUCTION.....                     | 3       |
|          | Background to the Problem.....        | 3       |
|          | Solution to the Problem.....          | 4       |
| 3.       | REQUEIREMNT SPECIFICATION.....        | 7       |
|          | 3.1 System Features.....              | 7       |
|          | 3.2 System Quality Attribute.....     | 9       |
|          | 3.3 System Interface.....             | 11      |
|          | 3.4 Project Requirements.....         | 12      |
| 4.       | FEATURES NOT TO BE TESTED.....        | 13      |
| 5.       | TESTING APPROACH.....                 | 14      |
|          | 5.1 Testing Levels.....               | 14      |
|          | 5.2 Test Tools.....                   | 15      |
|          | 5.3 Meetings.....                     | 17      |
| 6.       | TEST CASES/TEST ITEMS.....            | 18      |
| 7.       | TEST DELIVERABLES.....                | 33      |
| 8.       | STAFFING AND TRAINING NEEDS.....      | 33      |
| 9.       | TESTING SCHEDULE.....                 | 34      |
| 10.      | PLANNING RISKS AND CONTINGENCIES..... | 35      |
| 11.      | APPROVALS.....                        | 36      |

# **1. TEST PLAN IDENTIFIER: SQT-C04**

## **2. INTRODUCTION**

### **Background to the Problem**

Patients often experience difficulties when trying to obtain appointments through traditional methods. In many hospitals and clinics, appointments may be managed through phone calls, reception counters, or manual records. These approaches can become inefficient when there are many patients, multiple doctors, and limited appointment slots. As a result, patients may experience delays, confusion, and inconvenience during the appointment process.

One of the major problems is the difficulty of finding a suitable doctor and an available appointment slot. Patients may need information about a doctor's specialization, visiting schedule, and available appointment times before deciding. Collecting this information through traditional methods can require additional time and effort. Another problem is limited accessibility to appointment booking. Patients may need to contact or physically visit a hospital or clinic just to book an appointment or confirm a doctor's availability. This can be inconvenient for people who live far away or have limited time because of work, education, or other responsibilities.

Cancellation and rescheduling can also be difficult in a traditional appointment system. When a patient cannot attend a scheduled appointment, they may need to contact the hospital or clinic manually. If the cancellation or change is not properly recorded, the original appointment slot may remain unused. Doctors and administrative staff also face challenges when appointments, schedules, patient information, and cancellations are managed manually. Manual management can increase the possibility of human errors, duplicate bookings, and scheduling conflicts. It can also increase the workload of administrative staff.

Another important issue is the absence of effective appointment reminders. Patients may forget the date or time of their appointments, resulting in missed appointments and unused doctor slots. An automated reminder system can help patients keep track of their upcoming appointments and support better appointment management.

The security of patient information is also an important concern. An appointment system may contain personal information, doctor information, and appointment history. Therefore, the system needs to ensure that this information is properly protected from unauthorized access and poor record management.

The root cause of these appointment-related problems is the reliance on traditional and manually managed appointment procedures. When doctor information, availability, appointment bookings,

cancellations, rescheduling, and patient appointment records are managed through separate or manual methods, there is no centralized system to organize and update this information efficiently. This can make it difficult for patients to access accurate appointment information and can increase the possibility of scheduling errors, duplicate bookings, missed appointments, and unnecessary administrative work. The problem is important to consider because appointment management directly affects both patients and healthcare providers in several critical ways:

- **Patient Satisfaction and Access to Care:** When patients cannot easily find a suitable doctor or book an appointment at a convenient time, they may delay or avoid seeking medical care. This can negatively affect their health outcomes and overall satisfaction with healthcare services.
- **Healthcare Efficiency and Resource Utilization:** Missed appointments and unused doctor slots represent wasted healthcare resources. When appointment slots remain empty because of no-shows or poor cancellation management, other patients who need care may have to wait longer for appointments. This reduces the overall efficiency of healthcare delivery.
- **Administrative Burden and Costs:** Manual appointment management increases the workload on administrative staff, requiring them to spend time on phone calls, record-keeping, and conflict resolution. This can increase operational costs and reduce the time available for other important tasks.
- **Data Security and Privacy:** Patient information is sensitive and must be protected. A poorly managed appointment system may expose patient data to unauthorized access or loss. Ensuring data security is both a legal and ethical responsibility for healthcare providers.
- **Scalability and Growth:** As the number of patients and doctors grows, manual appointment systems become increasingly difficult to manage. A digital system can scale to accommodate more users without a proportional increase in administrative workload.

## **Solution to the Problem**

To address the problems mentioned above, the proposed solution is MediBook, an Online Doctor Appointment System is proposed that provides a centralized platform for managing doctor appointments. Patients can search for doctors based on specialization, location, and consultation fees, view available appointment slots, and book appointments online without making phone calls or visiting the clinic. They can also cancel or reschedule appointments when necessary. The system will send automated email and SMS reminders before appointments to help reduce missed appointments. Doctors can manage their schedules and upcoming appointments, while administrators can manage doctor profiles, view reports, and oversee the system. The system will also use role-based access control to protect patient information and ensure that sensitive data is only accessible to authorized users. This solution is particularly appropriate because it directly

addresses the root cause of the problems identified earlier. Instead of requiring patients to rely on phone calls or physical visits, the system provides all appointment-related information through one centralized digital platform. For example:

- A patient can search for a cardiologist, check available time slots, book an appointment, and receive confirmation within a few minutes.
- If the patient cannot attend, they can cancel or reschedule through the system, and the slot becomes available for someone else immediately.
- Doctors and administrative staff can manage schedules and appointments more efficiently, reducing human errors and workload.

The proposed solution is feasible and can meet the business objectives because the project has a clear and manageable scope focused specifically on appointment management, uses commonly available technologies like React.js, Node.js, and PostgreSQL, addresses a real and practical need as the problems identified affect both patients and healthcare providers, and is designed with scalability to accommodate future growth.

MediBook is a web-based platform designed to simplify the process of booking and managing doctor appointments. Its purpose is to provide a convenient and centralized way for patients, doctors, and administrators to handle appointments digitally. Patients can search for doctors, book, cancel, or reschedule appointments, and receive automated reminders. Doctors can manage their schedules and view upcoming appointments. Administrators can manage doctor profiles and generate reports. The main objectives of the system are to provide faster and more convenient appointment bookings, reduce scheduling errors and duplicate bookings, minimize missed appointments through automated reminders, reduce administrative workload, and ensure secure handling of patient information. By addressing these goals, the system aims to make healthcare appointments more organized, efficient, and accessible for everyone involved.

Online doctor appointment systems have become widely adopted worldwide. Studies show that digital appointment platforms reduce front-desk workload, minimize scheduling errors, and improve patient satisfaction. Below are some renowned platforms:

1. **Practo (India/Global):** Practo is one of the most trusted healthcare apps globally, allowing patients to search for doctors, view verified reviews, check real-time availability, and book appointments online. It also provides Practo Pro for doctors to manage appointments and send reminders. Practo operates in multiple countries and has facilitated over 100 million patient consultations.
2. **Zocdoc (USA):** Zocdoc is a leading healthcare marketplace in the United States that allows patients to search for in-network doctors by insurance and location, view real-time availability, and book appointments instantly online. It also offers virtual visit options.

#### Local Platforms (Bangladesh):

1. **DocTime:** DocTime is a leading healthcare app in Bangladesh that offers on-demand online doctor consultation via video/audio call/chat within 10 minutes. Patients can consult top doctors, book appointments, read real patient reviews, and choose suitable time slots.
2. **Maya:** Maya (formerly Maya Apa) is a Bangladesh-based digital wellbeing platform that connects users to a vetted network of experts including doctors, therapists, and lawyers. With approximately 10 million unique users, Maya helps people access stigma-free expert advice on sensitive topics.

These existing solutions show that digital appointment management is effective and widely accepted. However, the proposed project is focused on developing a simple Online Doctor Appointment System centered around appointment management, making it more manageable while still solving an important practical problem.

## **3. REQUIREMENT SPECIFICATION**

### **3.1 System Features**

#### **1. Patient Registration and Authentication**

- I. The system allows patients to register using their full name, valid email address, phone number, and a secure password.
- II. The system shall send a verification code (OTP) to the patient's registered email and phone number to confirm their identity before account activation.
- III. The system shall validate that the email address is in proper format and the phone number contains exactly 11 digits for Bangladeshi numbers.
- IV. The system allows patients to log in using their registered email and password to access their dashboard.
- V. If the username and/or password have been entered incorrectly for more than three consecutive attempts, the system shall generate and display a CAPTCHA verification code to retry login.
- VI. If the number of login attempts exceeds the limit (5 times), the system shall temporarily block the patient's account login for a period of two hours.

**Priority Level:** High

**Precondition:** System is connected to the database. Patients have a valid email and a phone number.

#### **2. Doctor Search and Discovery**

- I. The system shall allow patients to search for doctors by entering a specialization (e.g., Cardiologist, Dermatologist, Neurologist) in the search bar.
- II. The system allows patients to filter search results by location to find nearby healthcare providers.
- III. The system shall allow patients to filter search results by consultation fee range to find doctors within their budget.
- IV. The system allows patients to filter search results by doctor rating to find highly rated healthcare providers.
- V. The system allows patients to view complete doctor profiles including qualifications, years of experience, specialization, consultation fees, available time slots, and patient ratings.
- VI. The system shall display the doctor's available time slots in real-time based on their configured schedule, showing only slots that are not already booked.

**Priority Level:** High

**Precondition:** Patient is logged into the system. Doctor profiles exist in the database.

### 3. Appointment Booking and Management

- I. The system shall allow patients to select an available time slot from the doctor's schedule and book an appointment with a single click.
- II. The system shall prevent double booking by ensuring that a time slot cannot be booked by more than one patient simultaneously.
- III. The system shall display a confirmation message with appointment details (doctor name, date, time, location) immediately after successful booking.
- IV. The system shall send a confirmation email and SMS to the patient's registered email and phone number with complete appointment details.
- V. The system shall allow patients to cancel their upcoming appointments through their dashboard, releasing the time slot for other patients.
- VI. The system shall allow patients to reschedule their appointments by selecting a new available time slot from the same doctor's schedule.

**Priority Level:** High

**Precondition:** Patient is logged in and has selected a doctor and viewed available time slots.

### 4. Prescription and Medical Records Management

- i. The system shall allow doctors to upload electronic prescriptions for patients after consultation.
- ii. The system shall allow patients to view and download their prescriptions in PDF format.
- iii. The system shall allow patients to maintain a complete medical history including past illnesses, allergies, and current medications.
- iv. The system shall allow patients to upload medical documents and reports (e.g., lab reports, X-rays, diagnostic results).
- v. The system allows doctors to view the patient's medical history and previous prescriptions before consultation.
- vi. The system shall allow administrators to manage patient records and ensure data privacy compliance.

**Priority Level:** High

**Precondition:** Doctor is logged in and patient has completed an appointment.

### 5. Admin and Doctor Management Dashboard

- I. The system shall allow administrators to log in to a dedicated admin panel with enhanced privileges to manage the entire platform.

- II. The system shall allow administrators to add new doctors to the system by entering their name, specialization, license number, contact details, and consultation fees.
- III. The system shall allow administrators to edit or update doctor profiles including their schedules, fees, and contact information.
- IV. The system shall allow administrators to view all appointments in the system and generate reports including appointment history, cancellation rates, and revenue reports.
- V. The system shall allow doctors to log in to their dedicated dashboard to view their upcoming appointments, manage their availability schedule, and update their profile information.
- VI. The system shall allow doctors to mark appointments as "Completed" after the consultation, which updates the patient's appointment history and unlocks the ability for the patient to provide feedback.

**Priority Level:** High

**Precondition:** Admin and doctor accounts are created where proper authentication credentials are provided.

## 3.2 System Quality Attributes

### 1. Availability:

- The system must maintain a minimum uptime of 99.5% per month, especially during peak appointment booking hours (morning and evening).
- Planned maintenance windows must not exceed 4 hours per month and must be scheduled during off-peak hours (1:00 AM – 3:00 AM BST).

### 2. Performance:

- Doctor search results and available time slots must load and display within 3 seconds under normal load.
- The system must support a minimum of 2,000 concurrent active users without measurable performance degradation.
- Appointment booking confirmation (including database write and notification trigger) must be completed within 5 seconds.

### 3. Security (Integrity):

- All sensitive data, including patient NID numbers, medical records, and prescriptions, must be encrypted.
- Role-Based Access Control must be enforced so that patients, doctors, and administrators can only access authorized features and data.

- Two-factor authentication (OTP via SMS/email) must be applied during registration and after repeated failed login attempts.
- The platform must comply with the Bangladesh Digital Security Act, including audit trails for record access and user consent logging for medical-data sharing.

#### 4. Usability

- A first-time patient should be able to register, search for a doctor, and book an appointment within 8 minutes without external assistance.
- All critical workflows (login, doctor search, appointment booking, prescription download) must be accessible on devices with screen widths as small as 360px.
- Core patient actions (booking, cancelling, rescheduling) must be reachable within 3 clicks from the patient dashboard.

#### 5. Reliability

- The system must operate continuously for at least 3 consecutive months without a major failure requiring unplanned downtime.
- All data backups must be performed at least every 12 hours; data must be recoverable to within 2 hours of any failure point.

#### 6. Maintainability

- Critical bugs (severity: High), such as double-booking or failed report generation, must be diagnosed and patched within 72 hours of identification.
- All code modules must be documented with inline comments and API documentation so that a new developer can go onboard and contribute within 5 working days.

### 3.3 System Interface

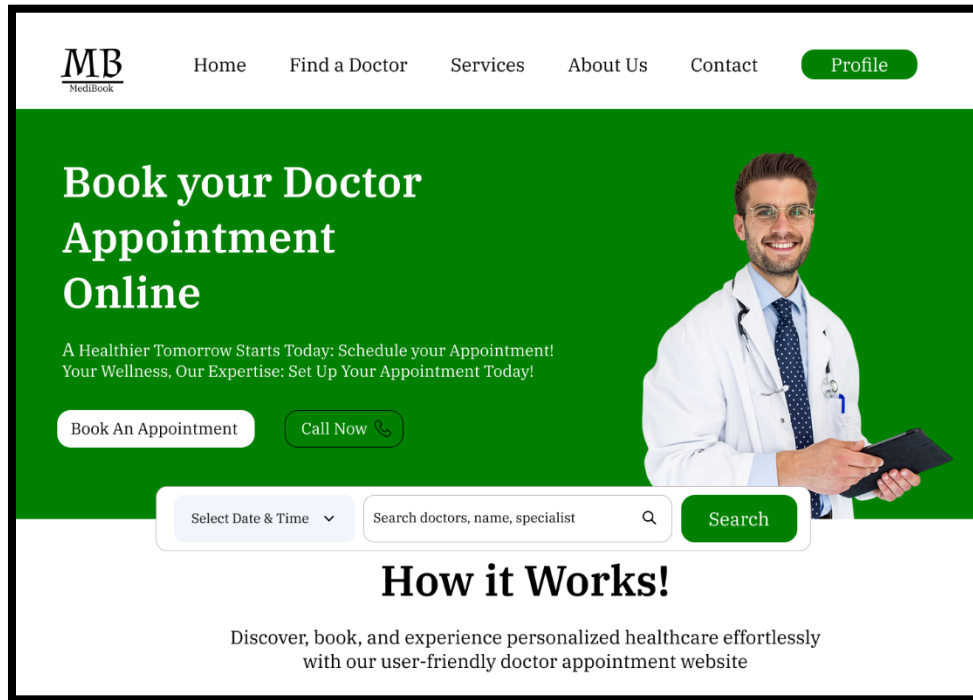


Figure-1: Home Page

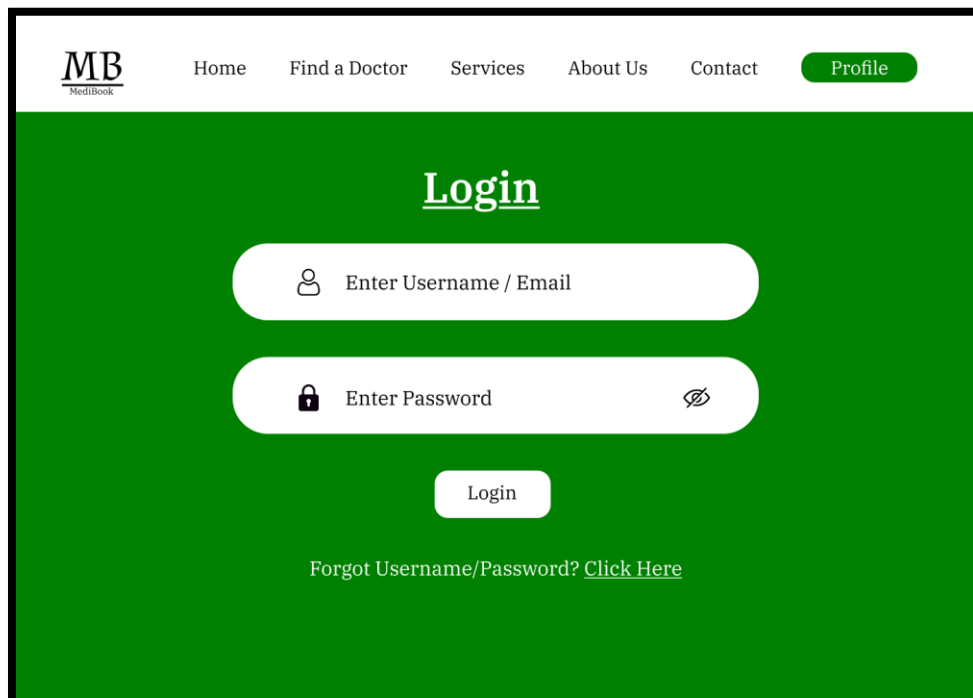


Figure-2: Login Page

The image shows a web browser window displaying the 'Patient Registration' page of a website called 'MediBook'. The website has a white header with the 'MB MediBook' logo on the left and navigation links: 'Home', 'Find a Doctor', 'Services', 'About Us', 'Contact', and a green 'Profile' button. The main content area has a green background with the title 'Patient Registration' in white. Below the title are six white input fields arranged in two columns: 'Enter Full Name...', 'Enter Email...', 'Enter Phone Number...', 'Enter Username...', 'Enter Password...', and 'Confirm Password...'. Below these fields is a checkbox labeled 'I agree to the Terms & Conditions'. At the bottom of the form is a white 'Create Account' button. Below the button is a link that says 'Already have an account? [Login](#)'.

Figure-3: New Patient Registration Page

### 3.4 Project Requirements

- Project Budget: 9,20,000 BDT
- Development Timeline: 3 Months (12 Weeks)
- Team Size: 5 Members (1 Frontend & UI/UX, 2 Backend, 1 QA, 1 PM)
- Tech Stack: React.js, Node.js, Express, PostgreSQL, bKash/Nagad Payment API

## **4. FEATURES NOT TO BE TESTED**

The following areas will not be specifically tested as part of the Online Doctor Appointment System. Any testing of these areas will be indirect and may occur because of testing the main system functionalities. These areas are considered outside the specific scope of this project:

1. **External Email and SMS Services:** The actual delivery performance, availability, and internal processing of third-party email and SMS services will not be tested. Testing will only verify that the system correctly sends the required notification requests.
2. **Payment Gateway Internals:** We will test if our system sends and receives data from the payment API, but we will not test the internal processing, security, or banking logic of the payment provider.
3. **Internet and Network Infrastructure:** The reliability, speed, and availability of the user's internet connection, network providers, or external network infrastructure will not be tested, as these are outside the control of the project.
4. **Third-Party Authentication Services:** The internal security, availability, reliability, and processing of third-party authentication or OTP services will not be tested. Testing will only verify that the Online Doctor Appointment System correctly sends authentication requests and processes the response received from the external service.
5. **Medical Decision Making:** The system is an appointment management platform. It does not diagnose diseases or provide medical treatment. Therefore, medical diagnosis and treatment decisions are outside the software testing scope.
6. **Doctor Profile Content Accuracy:** The factual accuracy of doctor qualifications, experience, and specialization details is the responsibility of the hospital administration and doctors. Testing only verifies that the system correctly stores and displays entered information.

## **5. TESTING APPROACH**

### **5.1 Testing Levels**

The testing for the Online Doctor Appointment System will consist of Unit, System/Integration, and Acceptance testing levels. It is preferred that at least one tester be involved in the system/integration testing phase. However, considering the project resources and timeline, testing will be performed collaboratively by the tester/test manager, development team, and selected end users.

UNIT Testing will be performed by the developers to verify individual functions and components of the system. Each developer will be responsible for testing the modules they implement, while the development team leader or test manager will review the test results. Unit testing will focus on the following modules:

- Patient Registration and Authentication Module: Testing email validation logic, password hashing functions, OTP generation and verification, and account lockout mechanisms.
- Doctor Search and Discovery Module: Testing search query processing, filter logic (location, fee range, rating), and real-time availability calculations.
- Appointment Booking and Management Module: Testing double-booking prevention logic, appointment confirmation workflow, cancellation, and rescheduling functions.
- Prescription and Medical Records Management Module: Testing document upload handling, file format validation (PDF), PDF generation, and medical record retrieval.
- Admin and Doctor Management Dashboard Module: Testing admin approval workflows, doctor profile management, and report generation functions.

SYSTEM/INTEGRATION Testing will be performed by the test manager and development team leader with assistance from the individual developers as required. No specific test tools are available for this project. Programs will enter System/Integration test after all critical defects have been corrected. A program may have up to two major defects if they do not impede testing of the program (i.e., there is a workaround for the error).

Key integration scenarios include:

- Registration Module ↔ Authentication Module: Verify that registration data flows correctly into the login system and that OTP verification completes the account activation process.
- Appointment Booking Module ↔ Notification Module: Verify that booking triggers confirmation emails and SMS to patients and notifications to doctors.
- Appointment Booking Module ↔ Doctor Schedule: Verify that booking an appointment correctly updates doctor's availability in real-time.
- Prescription Module ↔ Patient Profile: Verify that uploaded prescriptions are correctly linked to patient medical records and accessible from the patient dashboard.

- Admin Dashboard ↔ All Modules: Verify that admin actions (add doctor, approve accounts, generate reports) correctly update the system and reflect in user dashboards.
- Payment API ↔ Appointment Module: Verify that payment confirmation correctly updates appointment status.

ACCEPTANCE Testing will be performed by selected end users with assistance from the test manager and development team. Acceptance testing will determine whether the MediBook system satisfies the requirements and is suitable for actual use. This follows the course guidance that acceptance testing is formal testing performed by customers/end users to determine whether the system satisfies its acceptance criteria.

Acceptance testing will focus on complete real-world workflows, including:

- Patient Workflow: A patient registers an account, verifies the account using OTP, logs in, searches for a suitable doctor, checks available slots, books an appointment, receives confirmation, and later cancels or reschedules the appointment.
- Doctor Workflow: A doctor logs into the system, views upcoming appointments, manages availability, accesses relevant patient information, completes a consultation, and uploads an electronic prescription.
- Medical Record Workflow: A patient views their medical history, uploads medical reports, accesses a prescription uploaded by a doctor, and downloads the prescription in PDF format.
- Administrator Workflow: An administrator logs in, adds or updates doctor information, manages doctor schedules, views appointments, and generates required reports.
- End-to-End Workflow: A complete appointment process will be tested from patient registration through doctor selection, appointment booking, notification, consultation completion, prescription upload, and updating of the patient's medical record.
- Any defects or requirement mismatches identified during acceptance testing will be documented and reviewed before final approval of the system.

## 5.2 Test Tools

To provide appropriate testing coverage, the project will use a combination of manual and automated testing approaches:

### Manual Testing

- **Description:** Project team members will manually navigate the application and execute test cases to verify the system's functionality, usability, design, responsiveness, and overall

behavior. Manual testing will also be used for exploratory testing, complex edge cases, and scenarios that may not be easily detected through automated testing.

- **Associated Tasks:** Testing patient registration and OTP verification; checking doctor search, filtering, and profile information; verifying appointment booking, cancellation, rescheduling, and real-time slot availability; checking notification behavior; testing prescription and medical document upload/download workflows; verifying admin and doctor dashboards; testing UI responsiveness across different devices and screen sizes; performing exploratory testing on critical system functions; and checking the system's behavior when invalid, unexpected, or incomplete inputs are provided.

#### Automation Testing

- **Description:** Selenium WebDriver will be used to automate repetitive web-based functional and regression tests. Automation will help ensure that previously working features remain functional after new changes are introduced.
- **Associated Tasks:** Automating patient registration and login, testing CAPTCHA and login validation scenarios, automating doctor search and filtering, testing appointment booking workflows, verifying cancellation and rescheduling, and executing repeated regression tests after system updates.

#### Version Control

- **Tool: Git/GitHub**
- **Description:** Git/GitHub will be used for source-code version control and to maintain different versions of the project during development and testing.
- **Associated Tasks:** Code commits, branch management, merging changes, maintaining testing versions, and reviewing changes before they are included in the tested build.

### 5.3 Meetings

| Meeting Number | Date and Meeting Duration  | Reason                                                                                                     |
|----------------|----------------------------|------------------------------------------------------------------------------------------------------------|
| 1              | 25.07.2026<br>(1 hour)     | Review test cases, assign tasks, and prepare the test environment before execution begins.                 |
| 2              | 14.08.2026<br>(1 hour)     | Review test execution progress, analyze defect status, and address blockers to keep testing on schedule.   |
| 3              | 11.09.2026<br>(30 Minutes) | Review final test results, confirm all critical defects are resolved, and prepare for acceptance sign-off. |

## **6. TEST CASES/TEST ITEMS:**

Test Case-1:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                                                           |                                                                                                                                   | <b>Test Designed by:</b> Dipaneeta Pramanik                                                                                                          |                                                                         |                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------|
| <b>Test Case ID:</b> TC-1                                                                                                                                                 |                                                                                                                                   | <b>Test Designed date:</b> 27/07/26                                                                                                                  |                                                                         |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                                                            |                                                                                                                                   | <b>Test Executed by:</b> Dipaneeta Pramanik                                                                                                          |                                                                         |                    |
| <b>Module Name:</b> Patient Registration and Authentication                                                                                                               |                                                                                                                                   | <b>Test Execution date:</b> 29/07/26                                                                                                                 |                                                                         |                    |
| <b>Test Title:</b> Patient Registration with OTP Verification                                                                                                             |                                                                                                                                   |                                                                                                                                                      |                                                                         |                    |
| <b>Description:</b> Ensure patients can register using valid email and phone number.                                                                                      |                                                                                                                                   |                                                                                                                                                      |                                                                         |                    |
| <b>Precondition:</b> System is online and database is accessible                                                                                                          |                                                                                                                                   |                                                                                                                                                      |                                                                         |                    |
| Test Steps                                                                                                                                                                | Test Data                                                                                                                         | Expected Results                                                                                                                                     | Actual Results                                                          | Status (Pass/Fail) |
| 1. Go to The Website<br>2. Open The Registration Page<br>3. Enter Full Name<br>4. Enter Email Address<br>5. Enter Phone Number<br>6. Create Password<br>7. Click Register | Name: Rahima Jahan<br>Email: <a href="mailto:rahima@gmail.com">rahima@gmail.com</a><br>Phone: 01712345678<br>Password: Rahim@123@ | Patient registration should be completed successfully, and the system should send a verification code (OTP) to the registered email or phone number. | Registration was completed but OTP was not delivered by email or phone. | Fail               |
| <b>Post Condition:</b> Patient account created but cannot verify due to missing OTP.                                                                                      |                                                                                                                                   |                                                                                                                                                      |                                                                         |                    |

Test Case-2:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                                                                             |                                                                                                                               | <b>Test Designed by:</b> Dipaneeta Pramanik                                                                                                            |                |                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| <b>Test Case ID:</b> TC-2                                                                                                                                                                   |                                                                                                                               | <b>Test Designed date:</b> 30/07/26                                                                                                                    |                |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                                                                              |                                                                                                                               | <b>Test Executed by:</b> Dipaneeta Pramanik                                                                                                            |                |                    |
| <b>Module Name:</b> Patient Registration and Authentication                                                                                                                                 |                                                                                                                               | <b>Test Execution date:</b> 31/07/26                                                                                                                   |                |                    |
| <b>Test Title:</b> Block registration with invalid email and phone number formats                                                                                                           |                                                                                                                               |                                                                                                                                                        |                |                    |
| <b>Description:</b> Test whether the system rejects an invalid email address and phone number during registration.                                                                          |                                                                                                                               |                                                                                                                                                        |                |                    |
| <b>Precondition:</b> User is on the patient registration page                                                                                                                               |                                                                                                                               |                                                                                                                                                        |                |                    |
| Test Steps                                                                                                                                                                                  | Test Data                                                                                                                     | Expected Results                                                                                                                                       | Actual Results | Status (Pass/Fail) |
| 1. Go to the website<br>2. Open the registration page<br>3. Enter full name<br>4. Enter invalid email address<br>5. Enter an invalid phone number<br>6. Enter password<br>7. Click Register | Name: Rahima Jahan<br>Email: <a href="mailto:rahimamail.com">rahimamail.com</a><br>Phone: 01712345678<br>Password: Rahim@123@ | System should display error messages and should not allow registration until a valid email and exactly 11- digit Bangladeshi phone number are entered. | As expected    | Pass               |
| <b>Post Condition:</b> Patient’s account is not created.                                                                                                                                    |                                                                                                                               |                                                                                                                                                        |                |                    |

Test Case-3:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                                                         |                             |                                                                                                                                   | <b>Test Designed by:</b> Dipaneeta Pramanik |                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------|
| <b>Test Case ID:</b> TC-3                                                                                                                                               |                             |                                                                                                                                   | <b>Test Designed date:</b> 03/07/26         |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                                                          |                             |                                                                                                                                   | <b>Test Executed by:</b> Dipaneeta Pramanik |                    |
| <b>Module Name:</b> Patient Registration and Authentication                                                                                                             |                             |                                                                                                                                   | <b>Test Execution date:</b> 04/07/26        |                    |
| <b>Test Title:</b> Account Lockout on Failed Logins                                                                                                                     |                             |                                                                                                                                   |                                             |                    |
| <b>Description:</b> Verify account locks after 5 consecutive failed attempts.                                                                                           |                             |                                                                                                                                   |                                             |                    |
| <b>Precondition:</b> User account exists.                                                                                                                               |                             |                                                                                                                                   |                                             |                    |
| Test Steps                                                                                                                                                              | Test Data                   | Expected Results                                                                                                                  | Actual Results                              | Status (Pass/Fail) |
| 1. Go to the website<br>2. Open the login page<br>3. Enter wrong password (Attempt 1)<br>4. Repeat wrong password 5 times<br>5. Attempt 6th login with correct password | Password:<br>Wrong Password | The patient account locks for 2 hours and an “Account Locked” error message is displayed. Though CAPTCHA appears after 3 attempts | As expected                                 | Pass               |
| <b>Post Condition:</b> User cannot log in for 2 hours; security logs are updated.                                                                                       |                             |                                                                                                                                   |                                             |                    |

Test Case-4:

|                                                                                                                            |                              |                                                                   |                                                                              |                           |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------|
| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                            |                              |                                                                   | <b>Test Designed by:</b> Fahim Rahman Talha                                  |                           |
| <b>Test Case ID:</b> TC-4                                                                                                  |                              |                                                                   | <b>Test Designed date:</b> 05/08/26                                          |                           |
| <b>Test Priority (Low, Medium, High):</b> High                                                                             |                              |                                                                   | <b>Test Executed by:</b> Fahim Rahman Talha                                  |                           |
| <b>Module Name:</b> Doctor Search and Discovery                                                                            |                              |                                                                   | <b>Test Execution date:</b> 06/08/26                                         |                           |
| <b>Test Title:</b> Ensure patients can search for doctors by specialization                                                |                              |                                                                   |                                                                              |                           |
| <b>Description:</b> Test whether a patient can search for doctors using their specialization.                              |                              |                                                                   |                                                                              |                           |
| <b>Precondition:</b> Patient is logged in and doctor profiles are visible.                                                 |                              |                                                                   |                                                                              |                           |
| <b>Test Steps</b>                                                                                                          | <b>Test Data</b>             | <b>Expected Results</b>                                           | <b>Actual Results</b>                                                        | <b>Status (Pass/Fail)</b> |
| 1. Login to the system<br>2. Open the doctor search page<br>3. Enter a specialization in the search bar<br>4. Click Search | Specialization: Cardiologist | System should display doctors whose specialization is Cardiology. | System displayed no results even though cardiologists exist in the database. | Fail                      |
| <b>Post Condition:</b> Search results are not filtered correctly; patients cannot find doctors.                            |                              |                                                                   |                                                                              |                           |

Test Case-5:

|                                                                                                                                                    |                                           |                                                                                                                      |                                             |                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------|
| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                                    |                                           |                                                                                                                      | <b>Test Designed by:</b> Fahim Rahman Talha |                           |
| <b>Test Case ID:</b> TC-5                                                                                                                          |                                           |                                                                                                                      | <b>Test Designed date:</b> 07/08/26         |                           |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                                     |                                           |                                                                                                                      | <b>Test Executed by:</b> Fahim Rahman Talha |                           |
| <b>Module Name:</b> Doctor Search and Discovery                                                                                                    |                                           |                                                                                                                      | <b>Test Execution date:</b> 10/08/26        |                           |
| <b>Test Title:</b> Verify filtering doctors by location and consultation fee                                                                       |                                           |                                                                                                                      |                                             |                           |
| <b>Description:</b> Patient is logged in and doctor profiles with location and fee information exist.                                              |                                           |                                                                                                                      |                                             |                           |
| <b>Precondition:</b> Patient is logged in and doctor profiles are visible                                                                          |                                           |                                                                                                                      |                                             |                           |
| <b>Test Steps</b>                                                                                                                                  | <b>Test Data</b>                          | <b>Expected Results</b>                                                                                              | <b>Actual Results</b>                       | <b>Status (Pass/Fail)</b> |
| 1. Login to the system<br>2. Open doctor search<br>3. Enter or select location<br>4. Select consultation fee range<br>5. Click Search/Apply Filter | Location: Dhaka<br>Fee Range: 500–1000 BD | System should display doctors who are located in the Dhaka and whose consultation fees are within the selected range | As expected                                 | Pass                      |
| <b>Post Condition:</b> Filtered doctor results are displayed according to the selected criteria.                                                   |                                           |                                                                                                                      |                                             |                           |

Test Case-6:

|                                                                                                                       |  |                    |                                                                                                                                                                                           |                       |                           |
|-----------------------------------------------------------------------------------------------------------------------|--|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------|
| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                       |  |                    | <b>Test Designed by:</b> Fahim Rahman Talha                                                                                                                                               |                       |                           |
| <b>Test Case ID:</b> TC-6                                                                                             |  |                    | <b>Test Designed date:</b> 10/08/26                                                                                                                                                       |                       |                           |
| <b>Test Priority (Low, Medium, High):</b> High                                                                        |  |                    | <b>Test Executed by:</b> Fahim Rahman Talha                                                                                                                                               |                       |                           |
| <b>Module Name:</b> Doctor Search and Discovery                                                                       |  |                    | <b>Test Execution date:</b> 11/08/26                                                                                                                                                      |                       |                           |
| <b>Test Title:</b> View Doctor Profile Details                                                                        |  |                    |                                                                                                                                                                                           |                       |                           |
| <b>Description:</b> Test whether a patient can view complete doctor information                                       |  |                    |                                                                                                                                                                                           |                       |                           |
| <b>Precondition:</b> Patient is logged in and the selected doctor has a profile.                                      |  |                    |                                                                                                                                                                                           |                       |                           |
| <b>Test Steps</b>                                                                                                     |  | <b>Test Data</b>   | <b>Expected Results</b>                                                                                                                                                                   | <b>Actual Results</b> | <b>Status (Pass/Fail)</b> |
| 1. Login to the system<br>2. Search for a doctor<br>3. Select a doctor from the results<br>4. Open the doctor profile |  | Doctor: Akij Uddin | System should display the doctor's qualifications, experience, specialization, consultation fee, rating, and available time slots. Already booked slots should not be shown as available. | As expected           | Pass                      |
| <b>Post Condition:</b> User can view complete doctor information                                                      |  |                    |                                                                                                                                                                                           |                       |                           |

Test Case-7:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                                                            |  |                                                          | <b>Test Designed by:</b> Rayhan Chowdhury                                                                                                           |                |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| <b>Test Case ID:</b> TC-7                                                                                                                                                  |  |                                                          | <b>Test Designed date:</b> 12/08/26                                                                                                                 |                |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                                                             |  |                                                          | <b>Test Executed by:</b> Rayhan Chowdhury                                                                                                           |                |                    |
| <b>Module Name:</b> Appointment Booking and Management                                                                                                                     |  |                                                          | <b>Test Execution date:</b> 13/08/26                                                                                                                |                |                    |
| <b>Test Title:</b> Verify appointment booking with an available time slot.                                                                                                 |  |                                                          |                                                                                                                                                     |                |                    |
| <b>Description:</b> Ensure patients can book appointments successfully.                                                                                                    |  |                                                          |                                                                                                                                                     |                |                    |
| <b>Precondition:</b> Patient is logged in, a doctor has been selected, and an available time slot is displayed.                                                            |  |                                                          |                                                                                                                                                     |                |                    |
| Test Steps                                                                                                                                                                 |  | Test Data                                                | Expected Results                                                                                                                                    | Actual Results | Status (Pass/Fail) |
| 1.Login to the system<br>2. Search for a doctor<br>3. Select a doctor<br>4. View Doctor Profile<br>5. Check Available Slots<br>6. Select a time slot<br>7. Confirm Booking |  | Doctor: Dr. Rahman<br>Date: 30/08/2026<br>Time: 10:00 AM | Appointment should be successfully booked, and both email and sms confirmation message should display the doctor’s name, date, time, and location.. | As Expected    | Pass               |
| <b>Post Condition:</b> Appointment is saved in the database, and the selected time slot is no longer available to other patients. Doctor’s schedule is updated             |  |                                                          |                                                                                                                                                     |                |                    |

Test Case-8:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                              |  |                                                          | <b>Test Designed by:</b> Rayhan Chowdhury                                                             |                                                                                           |                    |
|------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------|
| <b>Test Case ID:</b> TC-8                                                                                                    |  |                                                          | <b>Test Designed date:</b> 14/08/26                                                                   |                                                                                           |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                               |  |                                                          | <b>Test Executed by:</b> Rayhan Chowdhury                                                             |                                                                                           |                    |
| <b>Module Name:</b> Appointment Booking and Management                                                                       |  |                                                          | <b>Test Execution date:</b> 17/08/26                                                                  |                                                                                           |                    |
| <b>Test Title:</b> Verify prevention of double booking                                                                       |  |                                                          |                                                                                                       |                                                                                           |                    |
| <b>Description:</b> Test whether the system prevents two patients from booking the same appointment slot.                    |  |                                                          |                                                                                                       |                                                                                           |                    |
| <b>Precondition:</b> The selected appointment slot has already been booked by another patient.                               |  |                                                          |                                                                                                       |                                                                                           |                    |
| Test Steps                                                                                                                   |  | Test Data                                                | Expected Results                                                                                      | Actual Results                                                                            | Status (Pass/Fail) |
| 1. Login as a patient<br>2. Search for the doctor<br>3. Select an already booked time slot<br>4. Try to book the appointment |  | Doctor: Dr. Rahman<br>Date: 30/08/2026<br>Time: 10:00 AM | System should prevent the second booking and display a message that the selected slot is unavailable. | System allowed the second booking for the same time slot. No error message was displayed. | Fail               |
| <b>Post Condition:</b> Double booking occurred; two patients booked the same slot                                            |  |                                                          |                                                                                                       |                                                                                           |                    |

Test Case-9:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                                                       |                                                 | <b>Test Designed by:</b> Rayhan Chowdhury                                                                                                                                              |                |                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| <b>Test Case ID:</b> TC-9                                                                                                                                             |                                                 | <b>Test Designed date:</b> 17/08/26                                                                                                                                                    |                |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                                                        |                                                 | <b>Test Executed by:</b> Rayhan Chowdhury                                                                                                                                              |                |                    |
| <b>Module Name:</b> Appointment Booking and Management                                                                                                                |                                                 | <b>Test Execution date:</b> 18/08/26                                                                                                                                                   |                |                    |
| <b>Test Title:</b> Verify appointment cancellation and release of time slot                                                                                           |                                                 |                                                                                                                                                                                        |                |                    |
| <b>Description:</b> Test whether a patient can cancel an upcoming appointment and make the time slot available again.                                                 |                                                 |                                                                                                                                                                                        |                |                    |
| <b>Precondition:</b> Patient has an upcoming appointment                                                                                                              |                                                 |                                                                                                                                                                                        |                |                    |
| Test Steps                                                                                                                                                            | Test Data                                       | Expected Results                                                                                                                                                                       | Actual Results | Status (Pass/Fail) |
| 1.Login to the system<br>2. Open My Appointments<br>3. Select an upcoming appointment<br>4. Click “Cancel Appointment” Button<br>5. Confirm cancellation<br>6. Verify | Appointment ID: APT001<br><br>Status: Confirmed | Appointment should be cancelled successfully and confirmation is displayed. The selected time slot should become available for other patients. Both cancellation email and sms is sent | As Expected    | Pass               |
| <b>Post Condition:</b> Appointment status is updated to be cancelled in the database, and the time slot is released.                                                  |                                                 |                                                                                                                                                                                        |                |                    |

Test Case-10:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                                                                                         |                                                                                                                                                    | <b>Test Designed by:</b> Mahin Sarker                                                                |                |                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------|--------------------|
| <b>Test Case ID:</b> TC-10                                                                                                                                                                              |                                                                                                                                                    | <b>Test Designed date:</b> 19/08/26                                                                  |                |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                                                                                          |                                                                                                                                                    | <b>Test Executed by:</b> Mahin Sarker                                                                |                |                    |
| <b>Module Name:</b> Prescription and Medical Records Management                                                                                                                                         |                                                                                                                                                    | <b>Test Execution date:</b> 20/08/26                                                                 |                |                    |
| <b>Test Title:</b> Verify doctor can upload an electronic prescription                                                                                                                                  |                                                                                                                                                    |                                                                                                      |                |                    |
| <b>Description:</b> Verify that doctors can upload electronic prescriptions for patients after completing a consultation                                                                                |                                                                                                                                                    |                                                                                                      |                |                    |
| <b>Precondition:</b> Doctor is logged in and the patient's appointment has been completed.                                                                                                              |                                                                                                                                                    |                                                                                                      |                |                    |
| Test Steps                                                                                                                                                                                              | Test Data                                                                                                                                          | Expected Results                                                                                     | Actual Results | Status (Pass/Fail) |
| 1. Login to the doctor’s dashboard.<br>2. Open the completed appointments section.<br>3. Select the patient.<br>4.Open the prescription section.<br>5. Upload the prescription file.<br>6. Click Upload | Doctor: Dr. Akij Mia<br>Patient: Rahima Jahan<br>Appointment ID: APT001<br>Prescription: Rahima_Prescription.pdf<br>File Format: PDF<br>Size: <5MB | The prescription should be uploaded successfully and linked to the correct patient's medical record. | As Expected    | Pass               |
| <b>Post Condition:</b> The prescription is stored in the patient's medical record and can be accessed by the authorized patient.                                                                        |                                                                                                                                                    |                                                                                                      |                |                    |

Test Case-11:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                                                                                                                  |                                                   | <b>Test Designed by:</b> Mahin Sarker                                                                       |                |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| <b>Test Case ID:</b> TC-11                                                                                                                                                                                                       |                                                   | <b>Test Designed date:</b> 21/08/26                                                                         |                |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                                                                                                                   |                                                   | <b>Test Executed by:</b> Mahin Sarker                                                                       |                |                    |
| <b>Module Name:</b> Prescription and Medical Records Management                                                                                                                                                                  |                                                   | <b>Test Execution date:</b> 24/08/26                                                                        |                |                    |
| <b>Test Title:</b> Doctor Views Patient Medical History Before Consultation                                                                                                                                                      |                                                   |                                                                                                             |                |                    |
| <b>Description:</b> Verify that doctors can view a patient's complete medical history and previous prescriptions before consultation                                                                                             |                                                   |                                                                                                             |                |                    |
| <b>Precondition:</b> Patient has an upcoming appointment, and patient has previous medical records                                                                                                                               |                                                   |                                                                                                             |                |                    |
| Test Steps                                                                                                                                                                                                                       | Test Data                                         | Expected Results                                                                                            | Actual Results | Status (Pass/Fail) |
| 1. Login to the doctor’s dashboard.<br>2. Navigate to "Upcoming Appointments"<br>3. Select a patient from upcoming list<br>4. Click "View Medical History"<br>5. View past prescriptions<br>6. View past illnesses and allergies | Doctor: Dr. Akij Mia<br><br>Patient: Rahima Jahan | Patient's complete medical history should be displayed.<br><br>Previous prescriptions are shown with dates. | As Expected    | Pass               |
| <b>Post Condition:</b> Doctor has full visibility of patient's medical history for better diagnosis                                                                                                                              |                                                   |                                                                                                             |                |                    |

Test Case-12:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                                                                              |                                                                                                                                                                                           | <b>Test Designed by:</b> Mahin Sarker                                                             |                |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------|--------------------|
| <b>Test Case ID:</b> TC-12                                                                                                                                                                   |                                                                                                                                                                                           | <b>Test Designed date:</b> 24/08/26                                                               |                |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                                                                               |                                                                                                                                                                                           | <b>Test Executed by:</b> Mahin Sarker                                                             |                |                    |
| <b>Module Name:</b> Prescription and Medical Records Management                                                                                                                              |                                                                                                                                                                                           | <b>Test Execution date:</b> 25/08/26                                                              |                |                    |
| <b>Test Title:</b> Patient Uploads Medical Reports and Documents                                                                                                                             |                                                                                                                                                                                           |                                                                                                   |                |                    |
| <b>Description:</b> Verify that patients can upload medical reports, lab results, and other health documents to their medical records                                                        |                                                                                                                                                                                           |                                                                                                   |                |                    |
| <b>Precondition:</b> Patient is logged in with a verified account                                                                                                                            |                                                                                                                                                                                           |                                                                                                   |                |                    |
| Test Steps                                                                                                                                                                                   | Test Data                                                                                                                                                                                 | Expected Results                                                                                  | Actual Results | Status (Pass/Fail) |
| 1. Login to the patient account.<br>2. Open Medical Records.<br>3. Select Upload Document.<br>4. Select a medical report.<br>5. Click Upload.<br>6. Verify uploaded document appears in list | Patient: Rahima Jahan<br>Document: Blood_Test_Report.pdf<br>Document Type: Laboratory Report<br>File: lab_report.pdf<br>Title: CBC Lab Report<br>Description: Complete blood count report | The medical document should be uploaded successfully and stored in the patient's medical records. | As Expected    | Pass               |
| <b>Post Condition:</b> The uploaded medical document is available in the patient's medical record.                                                                                           |                                                                                                                                                                                           |                                                                                                   |                |                    |

Test Case-13:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                 |                                                                                                                          | <b>Test Designed by:</b> Tanni Rani Nath                                                    |                |                    |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------|--------------------|
| <b>Test Case ID:</b> TC-13                                                                                                      |                                                                                                                          | <b>Test Designed date:</b> 25/08/26                                                         |                |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                  |                                                                                                                          | <b>Test Executed by:</b> Tanni Rani Nath                                                    |                |                    |
| <b>Module Name:</b> Admin Management Dashboard                                                                                  |                                                                                                                          | <b>Test Execution date:</b> 26/08/26                                                        |                |                    |
| <b>Test Title:</b> Verify admin can add a new doctor                                                                            |                                                                                                                          |                                                                                             |                |                    |
| <b>Description:</b> Administrator has a valid admin account and is logged in to the admin panel.                                |                                                                                                                          |                                                                                             |                |                    |
| <b>Precondition:</b> Patient has an upcoming appointment                                                                        |                                                                                                                          |                                                                                             |                |                    |
| Test Steps                                                                                                                      | Test Data                                                                                                                | Expected Results                                                                            | Actual Results | Status (Pass/Fail) |
| 1. Login to the admin panel<br>2. Open Doctor Management<br>3. Click Add Doctor<br>4. Enter doctor information<br>5. Click Save | Name: Dr. Sheikh Rahman<br>Specialization: Cardiologist<br>License No.: BMDC12345<br>Phone: 01712345678<br>Fee: 1000 BDT | New doctor’s profile should be successfully added to the system and stored in the database. | As Expected    | Pass               |
| <b>Post Condition:</b> Doctor profile is saved and can be viewed by patients in the doctor search section.                      |                                                                                                                          |                                                                                             |                |                    |

Test Case-14:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                 |                                                                  |                                                                                                            | <b>Test Designed by:</b> Tanni Rani Nath |                    |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------|
| <b>Test Case ID:</b> TC-14                                                                                                      |                                                                  |                                                                                                            | <b>Test Designed date:</b> 27/08/26      |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                  |                                                                  |                                                                                                            | <b>Test Executed by:</b> Tanni Rani Nath |                    |
| <b>Module Name:</b> Doctor Management Dashboard                                                                                 |                                                                  |                                                                                                            | <b>Test Execution date:</b> 28/08/26     |                    |
| <b>Test Title:</b> Verify doctor can manage availability schedule                                                               |                                                                  |                                                                                                            |                                          |                    |
| <b>Description:</b> Test whether a doctor can update their available schedule from the doctor dashboard.                        |                                                                  |                                                                                                            |                                          |                    |
| <b>Precondition:</b> Doctor has a valid account and is logged in to the doctor dashboard.                                       |                                                                  |                                                                                                            |                                          |                    |
| Test Steps                                                                                                                      | Test Data                                                        | Expected Results                                                                                           | Actual Results                           | Status (Pass/Fail) |
| 1. Login to the doctor dashboard<br>2. Open Schedule Management<br>3. Select a date<br>4. Enter available time<br>5. Click Save | Date:<br>01/09/2026<br>Available Time:<br>10:00 AM –<br>12:00 PM | Doctor's schedule should be successfully updated, and the available slots should be displayed to patients. | As Expected                              | Pass               |
| <b>Post Condition:</b> Updated schedule is stored in the database, and patients can see the available appointment slots.        |                                                                  |                                                                                                            |                                          |                    |

Test Case-15:

| <b>Project Name:</b> MediBook: Online Doctor Appointment System                                                                                                                                                                                                                                  |  |                                                             | <b>Test Designed by:</b> Tanni Rani Nath                                                  |                                                                           |                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------|
| <b>Test Case ID:</b> TC-15                                                                                                                                                                                                                                                                       |  |                                                             | <b>Test Designed date:</b> 31/08/26                                                       |                                                                           |                    |
| <b>Test Priority (Low, Medium, High):</b> High                                                                                                                                                                                                                                                   |  |                                                             | <b>Test Executed by:</b> Tanni Rani Nath                                                  |                                                                           |                    |
| <b>Module Name:</b> Admin and Doctor Management Dashboard                                                                                                                                                                                                                                        |  |                                                             | <b>Test Execution date:</b> 01/09/26                                                      |                                                                           |                    |
| <b>Test Title:</b> Verify admin can view appointment records and generate reports                                                                                                                                                                                                                |  |                                                             |                                                                                           |                                                                           |                    |
| <b>Description:</b> Test whether an administrator can view appointment information and generate appointment-related reports.                                                                                                                                                                     |  |                                                             |                                                                                           |                                                                           |                    |
| <b>Precondition:</b> Administrator has a valid admin account, is logged in, and appointment records exist in the database.                                                                                                                                                                       |  |                                                             |                                                                                           |                                                                           |                    |
| Test Steps                                                                                                                                                                                                                                                                                       |  | Test Data                                                   | Expected Results                                                                          | Actual Results                                                            | Status (Pass/Fail) |
| 1. Login to the admin panel<br>2. Open Appointment Management<br>3. View appointment records<br>4. Select the required report option<br>5. Generate the report1.<br>Login to the admin panel<br>2. Open Doctor Management<br>3. Click Add Doctor<br>4. Enter doctor information<br>5. Click Save |  | Report Type: Appointment History<br>Date Range: August 2026 | System should display the appointment records and generate the selected report correctly. | System displayed empty results or incorrect data in the generated report. | Fail               |

**Post Condition:** Administrator cannot view accurate appointment data.

## **7. TEST DELIVERABLES**

| Test Case ID | Test Title                                                     | Status |
|--------------|----------------------------------------------------------------|--------|
| TC-1         | Patient Registration with OTP Verification                     | Fail   |
| TC-2         | Block registration with invalid email and phone number formats | Pass   |
| TC-3         | Account Lockout on Failed Logins                               | Pass   |
| TC-4         | Ensure patients can search for doctors by specialization       | Fail   |
| TC-5         | Verify filtering doctors by location and consultation fee      | Pass   |
| TC-6         | View Doctor Profile Details                                    | Pass   |
| TC-7         | Verify appointment booking with an available time slot         | Pass   |
| TC-8         | Verify prevention of double booking                            | Fail   |
| TC-9         | Verify appointment cancellation and release of time slot       | Pass   |
| TC-10        | Verify doctor can upload an electronic prescription            | Pass   |
| TC-11        | Doctor Views Patient Medical History Before Consultation       | Pass   |
| TC-12        | Patient Uploads Medical Reports and Documents                  | Pass   |
| TC-13        | Verify admin can add a new doctor                              | Pass   |
| TC-14        | Verify doctor can manage availability schedule                 | Pass   |
| TC-15        | Verify admin can view appointment records and generate reports | Fail   |

## **8. STAFFING AND TRAINING NEEDS**

To provide complete and proper testing, the following training areas need to be addressed:

- Developers should be familiar with basic testing techniques such as unit testing, integration testing, and functional testing. They should also understand how to document test cases, expected results, and identify defects properly.
- Testers should be trained in test case execution, defect reporting, test documentation, and basic web application testing tools. If automation or performance testing tools such as Selenium WebDriver are used, testers should receive basic training on their use.
- Project team members should have a clear understanding of the main functions of the Online Doctor Appointment System, including patient registration, doctor search,

appointment booking, cancellation, rescheduling, notifications, and doctor/admin dashboards.

- Doctors and administrators should receive basic training on using their respective dashboards, managing doctor schedules, viewing appointments, managing doctor information, and generating reports.
- Selected end users should be trained in the basic use of the system before final acceptance. If direct training is not possible, a simple user manual or instructional video can be prepared to explain the main functions of the system.

## 9. TESTING SCHEDULE

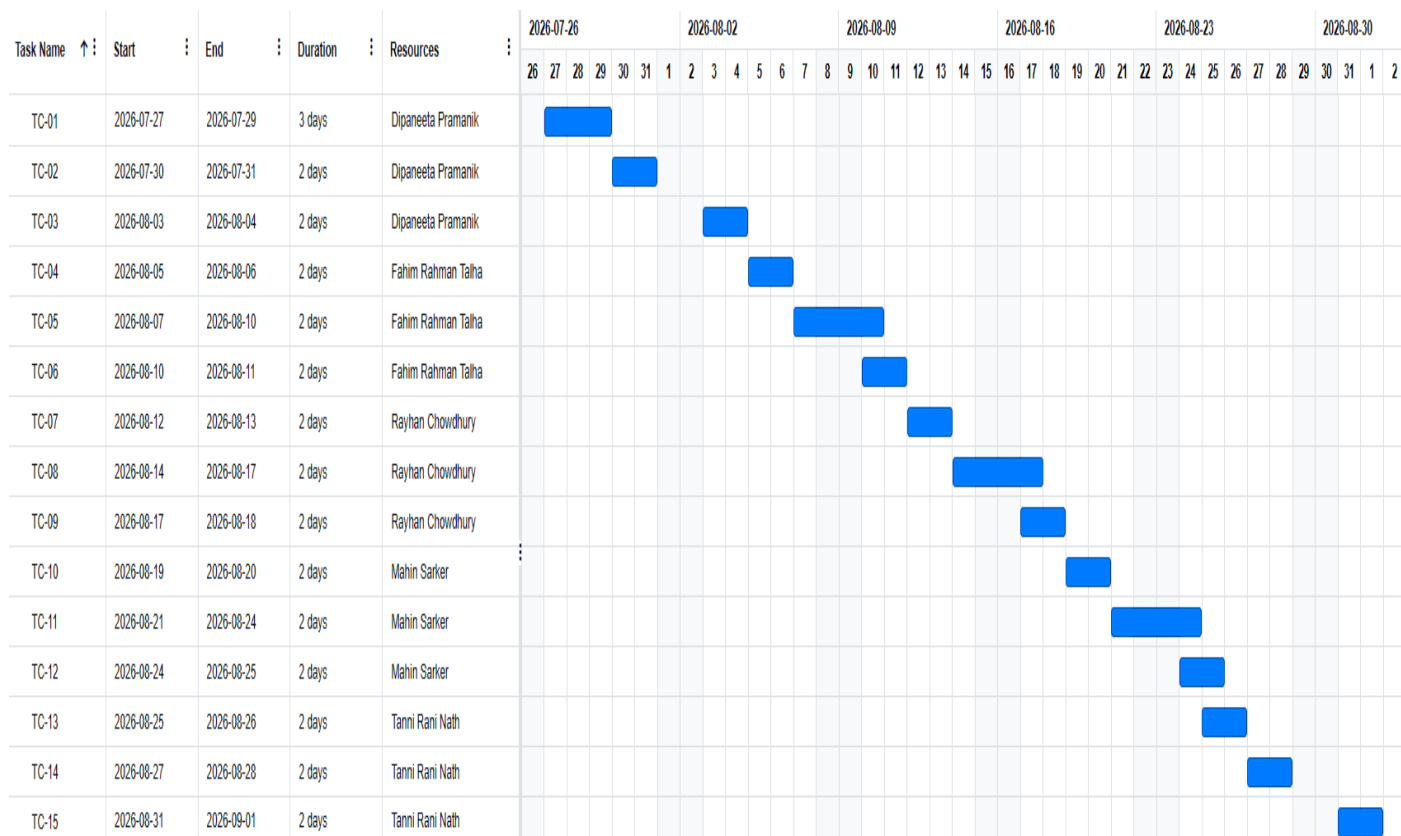


Figure-1: Gantt Chart for Testing Schedule.

## **10. PLANNING RISKS AND CONTINGENCIES**

| <b>Risk Identified</b>                       | <b>Severity</b> | <b>Mitigation Technique</b>                                                                                                                                                                                                                                        | <b>Impact</b>                                                                            | <b>Probability</b> |
|----------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------|
| Third-Party SMS/Email Service Downtime       | Medium          | Implement fallback notification mechanisms and retry logic. Maintain a queue of pending notifications to be sent once the service is restored.                                                                                                                     | Delayed or missed appointment confirmations and reminders                                | 35%                |
| Payment Gateway API Unavailability           | High            | Implement proper error handling and user-friendly error messages for failed transactions.                                                                                                                                                                          | Blocks payment and appointment confirmation testing                                      | 30%                |
| Doctor and Patient Unavailability for UAT    | Medium          | Schedule acceptance testing sessions in advance and coordinate with selected users. If some users are unavailable, trained project team members may perform representative test scenarios, while actual user validation will be completed before final acceptance. | Delays acceptance testing, feedback collection, and final system approval.               | 45%                |
| High Number of Defects During System Testing | High            | Prioritize critical and major defects for immediate correction. Maintain a defect-tracking process and reserve additional time before acceptance testing for regression and re-testing.                                                                            | Delays system testing may postpone the acceptance testing and final deployment schedule. | 55%                |
| Defect Fixing Delays                         | High            | Establish clear defect reporting and tracking system. Regular communication between teams; defect triage meetings; prioritize critical defects.                                                                                                                    | Blocked testing progress; delayed release                                                | 30%                |

|                                                    |        |                                                                                                                                                                                     |                                                                                                        |     |
|----------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----|
| Database or Server Failure During Testing          | Medium | Maintain regular database backups and a separate development/testing environment. Restore the environment when necessary and preserve test results and defect records separately.   | Interrupts testing activities and may cause loss of test data or test execution progress.              | 25% |
| Performance Problems During Peak Appointment Hours | High   | Conduct performance and load testing using Apache JMeter before acceptance testing. Identify slow database queries and optimize application and server performance where necessary. | Causes slow response times, failed requests, or an unsatisfactory user experience during busy periods. | 40% |

## **11. APROVALS**

|                                       |      |
|---------------------------------------|------|
| Sponsor                               | AIUB |
| Manager - Saikat Baul                 |      |
| Project Manager- Mahin Sarker         |      |
| Product Manager - Fahim Rahman Talha  |      |
| Development Manager- Rayhan Chowdhury |      |
| Test Manager - Tanni Rani Nath        |      |
| Account Manager -Dipaneeta Pramanik   |      |